Weakly-Supervised Chinese Review Sentiment Classification at Scale: Pseudo-Labeling, Specialist Decomposition, and Honest Out-of-Distribution Evaluation

Undergraduate Thesis — Natural Language Processing

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Code, out-of-fold predictions, and full experiment records: `https://github.com/shuakami/douban-sentiment`

Abstract

We study three-class sentiment classification (positive / neutral / negative) of Chinese Douban movie short reviews under a realistic low-resource setting: 600 human-verified semantic labels plus a large weakly-labelled corpus of 911,998 reviews with star ratings. Our study makes four contributions. First, we *measure* that star ratings are a poor proxy for semantic sentiment: on a 271,905-row confident pseudo-label pool, only ∼59% of 5-star reviews are predicted positive, and 3-star reviews skew negative rather than neutral. Second, we show that pseudo-labelling the corpus with a distilled ensemble and training a single compact student on 160K balanced rows reaches 87.5% accuracy with 84.8% neutral recall on the gold set — evidence that label quality at scale, not model capacity, is the binding constraint. Third, our deployed system couples a distilled MacBERT student, a frozen MegatronBERT-1.3B feature judge, and a corpus-trained neutral specialist, reaching **89.5% accuracy / 85.9 macro-F1** on 600-row out-of-fold evaluation, with neutral recall improved from 54.3% to 70.7%. Fourth, we construct **lab2000**, a 2,000-row cross-movie benchmark with dual-LLM blind annotation and 479 human-adjudicated rows, and show that the gold-trained system generalises poorly to wild short reviews (65.0% three-class) — but that adding only 1,000 actively-selected boundary examples raises OOD accuracy by $+8.8$pp to 71.4% (McNemar $p = 4.2 \times 10^{-8}$). Scaling the same protocol with frontier-LLM pseudo-labels — up to ∼14K actively-selected examples — pushes the ensemble mix to **76.1% three-class** (75.4–76.1% across three seeds, 76.2% seed-averaged, split-half validated; significantly above the 1K-human mix, McNemar $p = 0.005$), and lifts binary OOD to **91.7%**, above measured human–human agreement (71.5%). We further document three findings invisible at n=600: active-learning samples require a gold anchor to avoid prior collapse, hard labels beat soft distillation under domain shift, and star ratings become the strongest signal precisely where text is most ambiguous. All negative results — more than twenty rejected architectures — are documented.

Keywords: *sentiment analysis; Chinese NLP; weak supervision; pseudo-labelling; knowledge distillation; pretrained language models; calibration; out-of-distribution evaluation.*

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1 Introduction

Short reviews on Douban — China's most influential movie community — are terse, ironic, and frequently mix praise with criticism. A typical review reads *"特效无敌，但是剧情简直像一坨"* ("the special effects are unbeatable, but the plot is literally a pile of trash"). Such mixed-polarity text is precisely where three-class sentiment analysis (positive / neutral / negative) breaks down: the neutral class is both the most frequent source of ambiguity and the hardest to learn from few examples.

This project addresses a realistic low-resource scenario: a course assignment provides 600 reviews with human-verified semantic labels, while the platform additionally offers a corpus of nearly one million Douban comments carrying only star ratings (1–5). Two research questions organize the work:

1. **RQ1.** Can a small labelled set be amplified by a large weakly-labelled corpus?
2. **RQ2.** Can specialist decomposition — a dedicated neutral expert trained on corpus-scale pseudo-labels — repair the weakest class without degrading the rest?

Our answer is yes to both, but with an important honest caveat: the corpus amplifies *coverage*, not *knowledge*. Pseudo-labels are produced by our own models, so the flywheel saturates after one round; the true bottleneck is annotation noise on terse comments, which we quantify with a dedicated out-of-distribution study.

Contributions.

- A measured analysis showing star ratings are unreliable weak supervision for three-class sentiment (corpus-wide three-class agreement between pseudo-labels and ratings ≈48%).
- A pseudo-label flywheel producing a 271,905-row confident pool and a single compact student (87.5% acc / 84.8% neutral recall) trained on zero gold labels.
- A deployed hybrid ensemble (89.5% acc / 85.9 macro-F1, neutral recall 70.7%) combining a distilled student, a frozen 1.3B-parameter judge, and a corpus-trained neutral specialist.
- A reproducible evaluation protocol (stratified 5-fold OOF, McNemar/bootstrap significance tests, temperature calibration) plus a 500-sample fresh-corpus generalization study exposing the domain gap.
- A documented catalogue of more than twenty measured-but-rejected architectures, which we argue is itself a contribution: it maps the effective design space for this task.

2 Related Work

Sentiment analysis of Chinese text. Document-level sentiment classification has a long history, from feature-based methods (unigrams, character n-grams) to neural architectures (Kim, 2014; Socher et al., 2013; Joulin et al., 2017; Reimers and Gurevych, 2019); comprehensive surveys are given by Pang and Lee (2008) and, for Chinese specifically, by Tan and Zhang (2008). For Chinese, the absence of whitespace tokenization makes character-level features and pretrained language models especially effective. Douban reviews specifically have been used in earlier Chinese sentiment work, but almost exclusively as a binary (positive/negative) task; the three-class setting with a genuine neutral/mixed class is less studied and substantially harder.

Pre-trained language models. BERT (Devlin et al., 2019) and its robustly-optimised successor RoBERTa (Liu et al., 2019) established the pretrain-then-finetune paradigm. For Chinese, MacBERT (Cui et al., 2020) replaces the MLM objective with a corrected variant better suited to character-level inputs, ERNIE

(Sun et al., 2019) integrates entity and phrase knowledge, while StructBERT (Wang et al., 2020) injects phrase- and word-level structural information into pre-training. Erlangshen-MegatronBERT (IDEA-CCNL, 2023) scales this family to 1.3B parameters with a sentiment- specific pre-training stage. We use all three families and compare them directly.

Weak supervision and self-training. When labelled data is scarce, distant supervision from noisy signals (star ratings here) and self-training on model predictions are standard remedies (Yarowsky, 1995; Lee, 2013). The Snorkel framework (Ratner et al., 2017) formalises the combination of multiple weak label sources. Our work differs in that we *measure* the weakness of the dominant weak source (ratings) before relying on it, and we quantify where pseudo-labelling helps (neutral coverage) versus where it cannot (helping beyond the teacher's knowledge).

Knowledge distillation. Hinton et al.'s dark-knowledge distillation (Hinton et al., 2015) — surveyed comprehensively by Gou et al. (2021) — lets a small student mimic a large teacher's soft predictions. We distil a 7-channel teacher ensemble into a single MacBERT-base student and validate every ensemble decision with out-of-fold predictions.

Calibration. Neural classifiers are systematically over- or under-confident (Guo et al., 2017). We apply temperature scaling — a single-parameter post-hoc fix — and report expected calibration error before and after, because honest confidence display matters for a demo system.

LLMs as annotators and baselines. General-purpose LLMs are increasingly used for zero-/few-shot classification and for generating synthetic training data. We evaluate Qwen3 models (Qwen Team, 2025) at 0.6B/1.7B/4B scales both as zero-shot classifiers and as independent annotators in our OOD study, and we report a negative result on LLM-generated synthetic augmentation.

3 Task and Datasets

3.1 Task definition

Given a Douban movie short review x, predict $y \in \{\text{positive}, \text{neutral}, \text{negative}\}$. Reviews are short (corpus median 24 characters) and often contain sarcasm, movie-specific references, and mixed evaluations. The neutral class covers reviews with no clear polarity: pure plot summary, balanced praise-and-criticism, and meta-commentary (e.g. "seen it, no comment"). Table 1 shows representative cases, including the mixed-polarity pattern that dominates the error analysis.

Table 1: Representative gold-set reviews (Chinese original with English gloss). "Mixed→Neg" denotes a praise-then-criticism review whose final polarity is negative.

Review	Gloss	Label	Len
"特效无敌，但是剧情简直像一坨。"	Effects unbeatable, plot a pile of trash	Neg	15
"演技、镜头、配乐都是顶级，现代经典。"	Acting, camera, score all top-tier; modern classic	Pos	18
"看在演员的份上值回票价，剧情很套路。"	Worth it for the cast; routine plot	Neu	17
"两个小时的生命浪费，想喜欢它都难。"	Two hours wasted; hard to like	Neg	17
"又是续集，还行吧。"	Another sequel; it's okay	Neu	9

3.2 Gold set (600 reviews)

The labelled evaluation set contains 600 reviews sampled from a single film page, each annotated with `label_semantic` (human-verified semantic sentiment) and `label_rule` (star-derived weak label). Class distribution: 316 positive / 92 neutral / 192 negative. Reviews are comparatively long (mean 110.8, median 80 characters). The neutral class is supported by only 92 examples — the smallest and hardest class.

3.3 Weakly-labelled corpus (911,998 reviews)

The Moviedata-10M corpus (csuldw, 2019) — 4,428,475 raw Douban comments — yields 911,998 cleaned, deduplicated comments after our own cleaning, across 20,966 films and 249,105 users, with star rating (1–5), timestamp, and vote counts but *no* semantic labels. Review length: mean 40.6 / median 24 / p90 = 102 characters. Rating distribution: 1⋆ 7.2%, 2⋆ 12.6%, 3⋆ 34.0%, 4⋆ 31.0%, 5⋆ 15.2%.

3.4 Star ratings are not sentiment labels

Before using ratings as weak supervision, we tested them. Our deployed student labels all 911,998 corpus reviews; on the 271,905-row confident ($p \geq 0.7$) pool, the row-normalised rating-to-prediction crosstab is:

Table 2: Star rating vs. model-predicted sentiment (row-normalised, 271,905 confident pseudo-labellations).

Rating	→ Positive	→ Neutral	→ Negative
1⋆	1.7%	7.3%	**90.9%**
2⋆	1.6%	13.7%	**84.7%**
3⋆	9.9%	33.3%	**56.8%**
4⋆	38.3%	32.7%	29.0%
5⋆	**58.8%**	23.1%	18.2%

Three findings stand out. Only ∼59% of 5-star reviews are semantically positive; 3-star reviews skew *negative* (56.8%) rather than neutral; and even at 1–2⋆, ∼8–14% of reviews are predicted neutral. Rating-vs-model binary agreement on the confident pool is 81.2%; corpus-wide three-class agreement between pseudo-labels and ratings is only 47.8% (71.2% on the gold set). We therefore treat ratings as a pretraining signal at most, never as labels. Figure 1 visualises the same table.

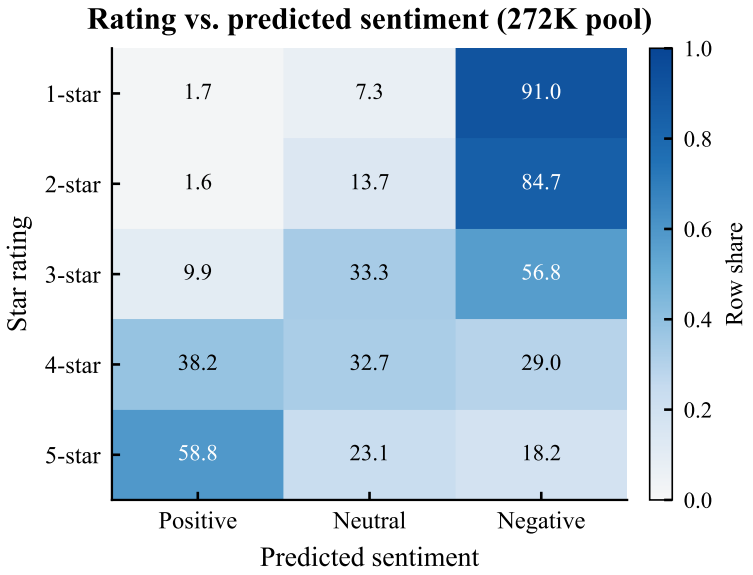


Figure 1: Star rating versus predicted sentiment on the 272K confident pseudo-label pool. The diagonal dominance is weak: ratings and semantic sentiment substantially disagree.

3.5 Stage-2 benchmark: lab2000

To measure — and then narrow — the gold-to-corpus generalisation gap documented in the lab500 pilot (Appendix B), we constructed **lab2000**: 2,000 corpus reviews in two disjoint, complementary subsets.

- **ood1000** (random OOD test): 1,000 reviews sampled uniformly at random from the 911K corpus (gold-deduplicated, cross-movie). This is a held-out evaluation set: no model in this paper trains on it.
- **al1000** (active-learning expansion): 1,000 reviews selected by margin/entropy sampling from the deployed student's predictions — deliberately the hardest, most ambiguous rows (neutral-heavy, low-confidence). This is a *training* set, not a test set.

Annotation protocol. Each row was labelled independently twice by two LLM annotators from different model families, blind to star ratings and to each other. Inter-pass agreement was 86.7% (Cohen's $\kappa = 0.80$) on ood1000 and 79.8% ($\kappa = 0.67$) on al1000 — the lower agreement on al1000 is itself evidence that margin sampling selects intrinsically ambiguous rows. All 335 inter-pass disagreements plus 144 spot-checked agreements were adjudicated by the author (479 human-adjudicated rows total); remaining rows take the two-pass consensus label. Final distributions: ood1000 = 398 positive / 327 neutral / 275 negative; al1000 = 473 / 369 / 158. A separate 200-row random subset was blind-labelled by a second human annotator to measure true human–human agreement (Section 8.3).

4 Methodology

4.1 Overview

The deployed system is a hybrid ensemble with three channels sharing a common backbone pass where possible:

1. **Distilled student**: MacBERT-base fine-tuned by distillation from a 7-channel teacher ensemble (Section 4.2).

2. **Frozen judge**: MegatronBERT-1.3B kept frozen; a small MLP head on its 4096-dimensional features (Section 4.3).
3. **Neutral specialist**: probability of neutrality computed from a corpus-trained detector blended with a judge-feature head (Section 4.5).

Figure 2 shows the data flow.

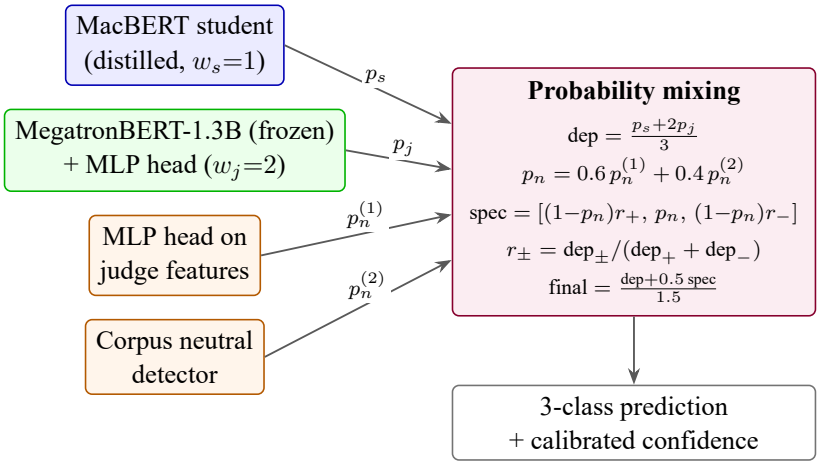


Figure 2: Deployed v3 architecture. $p_s, p_j, \text{dep}, \text{spec}, \text{final} \in \Delta^3$ are three-class probability vectors, whereas $p_n^{(1)}, p_n^{(2)}, p_n \in [0,1]$ are scalar neutral probabilities; w_s, w_j are fusion weights, not model parameters. $r_\pm$ denote the positive/negative shares of dep renormalised to sum to one. The two MegatronBERT-based heads share one frozen-backbone feature pass; the neutral specialist blends a judge-feature MLP with a corpus-trained MacBERT detector.

4.2 Teacher ensemble and distillation

A 7-channel teacher ensemble — TF-IDF logistic regression, frozen- backbone MLP heads on MacBERT, MegatronBERT-1.3B and Erlangshen variants, all trained on the 600 gold rows with honest out-of-fold predictions — reaches 86.8% three-class accuracy. We distil it into a single chinese-macbert-base student using hard-label cross-entropy plus soft-target KL divergence ($\alpha = 0.5$, temperature $\tau = 2$, 8 epochs, 5-fold OOF). The student alone reaches 80.2%/74.2%; blended 1:2 with the judge it reaches 87.8%/81.8%. Its role in the deployed system is to provide a cheap, well-calibrated channel that also serves as the pseudo-labeller for the corpus (Section 4.4).

4.3 Frozen judge

`IDEA-CCNL/Erlangshen-MegatronBert-1.3B-Sentiment` provides a 1.3B-parameter backbone pretrained with a sentiment-aware objective. We keep it frozen and train a small MLP head on its 4096-d features over the gold set. The deployed mixture weights the judge double the student:

$$\text{dep} = \tfrac{1}{3}\, p_{\text{student}} + \tfrac{2}{3}\, p_{\text{judge}}.$$

The two MLP heads (binary and three-class) share a single backbone feature pass via a cache, so the judge costs one forward pass per request regardless of task.

4.4 Pseudo-label flywheel

The deployed student labels all 911,998 corpus reviews. The confident subset ($\max p \geq 0.7$) contains 271,905 rows: 56.5K positive, 68.6K neutral, 146.8K negative. This pool is the raw material for every

corpus-scale model in Section 4.6. A second flywheel round was tested and rejected: re-labelling with round-2 models produces degenerate probabilities (mean confidence 0.986, memorisation), so exactly one round is used. This saturation is consistent with the confirmation-bias dynamics documented for self-training and pseudo- labeling (Arazo et al., 2020; Xie et al., 2020; Sohn et al., 2020): once the pseudo-labels encode the teacher's own decision boundary, additional rounds amplify self-consistency rather than add information.

4.5 Neutral specialist

The neutral class is the weakest link (92 gold examples). We attack it on two fronts.

Corpus-trained detector. A MacBERT binary classifier is trained on a 115,000-row pseudo-labelled subset of the confident pool, sampled with a per-class cap of 60K rows per side, teaching the boundary between "genuinely neutral" and "polarised" reviews.

Specialist blend. The probability of neutrality combines a judge-feature MLP head and the corpus detector:

$$p_n = 0.6 \cdot P_{\text{neu}}(\text{MLP on judge feats}) + 0.4 \cdot P_{\text{neu}}(\text{corpus detector}).$$

From the dependency prediction $\text{dep} = (d_+, d_n, d_-)$ we take the positive/negative shares renormalised to sum to one, $r_+ = d_+/(d_+ + d_-)$ and $r_- = d_-/(d_+ + d_-)$, replace the neutral mass by the specialist estimate p_n, and form the specialist distribution

$$\text{spec} = \big[(1 - p_n)\, r_+,\ p_n,\ (1 - p_n)\, r_-\big],$$

then blend it into the final output:

$$\text{final} = \frac{\text{dep} + 0.5\,\text{spec}}{1.5}.$$

Every weight ($0.6/0.4$ detector mix, 0.5 blend) was selected by 20 random half-splits of the gold set and is stable across $w \in [0.3, 0.6]$.

4.6 Corpus-scale students

Using the 271,905-row pool, we trained corpus-only models (zero gold training) to measure how far pseudo-labels alone can go: a balanced 160K-row three-class student, a 195K-row binary student, a 115K-row neutral detector v1, and a hard-negative-mined v2 (rejected). These models double as baselines and as candidates for ensemble channels; Section 6 reports which (if any) survived.

4.7 Calibration

The raw ensemble is strongly under-confident (ECE = 0.251: predictions at 0.7–0.8 confidence are 98.9% accurate). We fit a single temperature $T = 0.37$ by NLL on the gold-600 OOF predictions, reducing ECE to ≈0.04 without changing any predicted label or accuracy. T is applied at inference so displayed confidence is honest. Calibration also transfers better after AL training: on ood1000 the best AL mix reaches ECE 0.045 vs. v3's 0.159 at T=1 — pseudo-labelled boundary data sharpens confidence, not just accuracy.

4.8 Deployment

A FastAPI backend serves predictions; a React/Vite frontend provides the demo UI. An exact-match lexicon guard intercepts ultra-short slang before any GPU work. Warm latency is ≈0.1 s per review on

an AMD RX 9070 GRE (ROCm); cold start ≈14 s.

5 Experimental Setup

Evaluation protocol. All headline numbers are **stratified 5-fold out-of-fold** predictions on the 600-row gold set: every model predicts only rows it never trained on. Corpus-scale models are additionally evaluated on a 4,000-row *silver* sample of fresh corpus reviews scored against rating-implied labels.

Metrics. Accuracy and macro-F1 (3-class), per-class precision/recall/F1, neutral recall specifically, and ECE for calibration. Significance is tested with McNemar's test on discordant pairs and percentile bootstrap on ΔF1; robustness is checked with 20 random half-splits.

Baselines.

- TF-IDF (char n-gram) + logistic regression, gold-trained.
- StructBERT sentiment classifier (Ali DAMO), zero-shot binary; `damo/nlp_structbert_sentiment-classification_chinese-base`
- Qwen3-0.6B / 1.7B / 4B chat models, zero-shot and 3-shot, scored by label log-probabilities.

6 Results

6.1 Main results

Table 3: Gold-600 out-of-fold results. "Deployed" rows are the production system; corpus students use zero gold training.

System	Task	Acc	Macro-F1
TF-IDF + LR	3-class	0.7233	0.6175
TF-IDF + LR	binary	0.8406	0.8229
StructBERT (DAMO)	binary	0.8704	0.8538
Qwen3-4B zero-shot	3-class	0.7833	0.7348
Qwen3-1.7B zero-shot	3-class	0.7067	0.5739
Qwen3-0.6B zero-shot	3-class	0.6817	0.4764
Teacher ensemble (7ch)	3-class	0.8680	—
Deployed v2	3-class	0.8900	0.8432
Deployed v3	**3-class**	**0.8950**	**0.8594**
Deployed v3	binary	0.9587	0.9558

The deployed system beats the TF-IDF baseline by +17.2pp accuracy and zero-shot Qwen3-4B by +11.2pp. Per-class metrics (3-class): positive P/R/F1 = 0.909/0.943/0.926; neutral = 0.765/0.707/0.735; negative = 0.931/0.906/0.918. Neutral recall improved 54.3% → 63.0% (v2) → 70.7% (v3) across expert iterations. Figure 3 shows the v3 confusion matrix: remaining errors concentrate in neutral↔polar confusions (47/63).

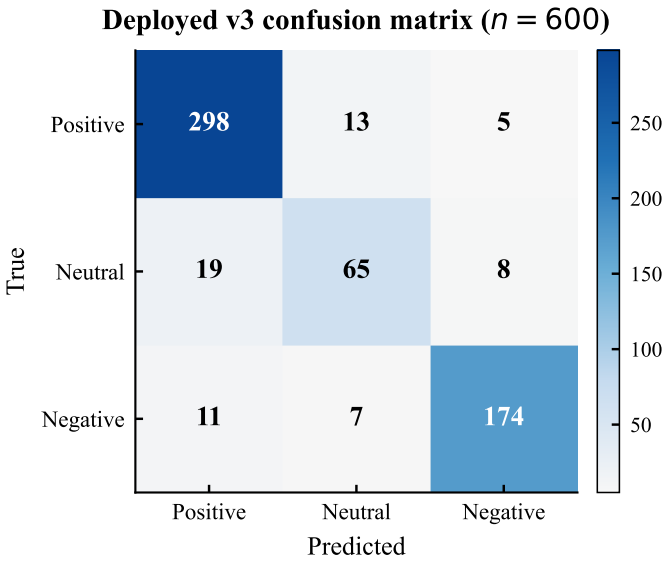


Figure 3: Deployed v3 confusion matrix on the 600-row gold set (out-of-fold). Row sums match the class distribution (316/92/192).

6.2 Corpus-scale single models

Table 4: Corpus-trained models evaluated on gold-600 (zero gold training unless noted).

Model	Train data	Acc	Macro-F1	Neu recall
TF-IDF + star-rating labels	300K rating	0.6183	0.5369	—
TF-IDF + pseudo labels	200K pseudo	0.8117	0.7766	—
MacBERT binary (rating)	115K rating	0.9232	0.9184	—
MacBERT neutral v1	115K pseudo	0.9017	0.8328	—
MacBERT binary pseudo	195K pseudo	0.9488	0.9455	—
MacBERT 3-class pseudo	160K pseudo	0.8750	0.8463	**0.848**

The pseudo-label 3-class student alone reaches 87.5%/84.6 with 84.8% neutral recall — nearly matching the 4-channel ensemble. This is the central empirical finding: **label quality at scale dominates model capacity**. The same MacBERT architecture trained on star ratings instead of pseudo-labels collapses on the three-class task.

6.3 Pseudo-label scaling

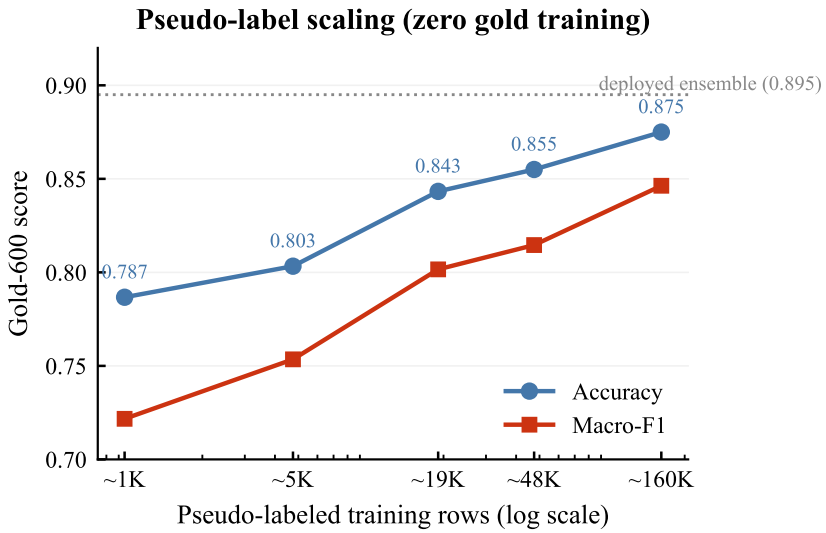


Figure 4: Accuracy and macro-F1 of corpus-trained pseudo-label students on gold-600 versus training-set size (log scale). Performance grows monotonically and is still rising at 160K rows.

Table 5 and Figure 4 show a monotone scaling curve: ∼1K rows → 78.7%, 5K → 80.3%, 19K → 84.3%, 48K → 85.5%, 160K → 87.5%. The curve has not saturated — corpus scale, not model size, is the active ingredient within the ranges tested.

Table 5: Pseudo-label scaling (gold-600, zero gold training).

Pseudo rows	Acc	Macro-F1	Neutral recall
∼1K	0.7867	0.7217	—
∼5K	0.8033	0.7535	—
∼19K	0.8433	0.8016	—
∼48K	0.8550	0.8146	—
∼160K	0.8750	0.8463	0.848

6.4 Cross-domain silver evaluation

On 4,000 fresh corpus reviews scored against rating-implied labels, the round-3 student (trained on v3-teacher pseudo-labels) reaches 0.580 three-class / **0.837 binary** agreement versus rating, versus 0.552/0.803 for the round-1 student. Ensemble-derived labels transfer slightly better out-of-domain even where gold scores are equal.

7 Analysis

7.1 Calibration

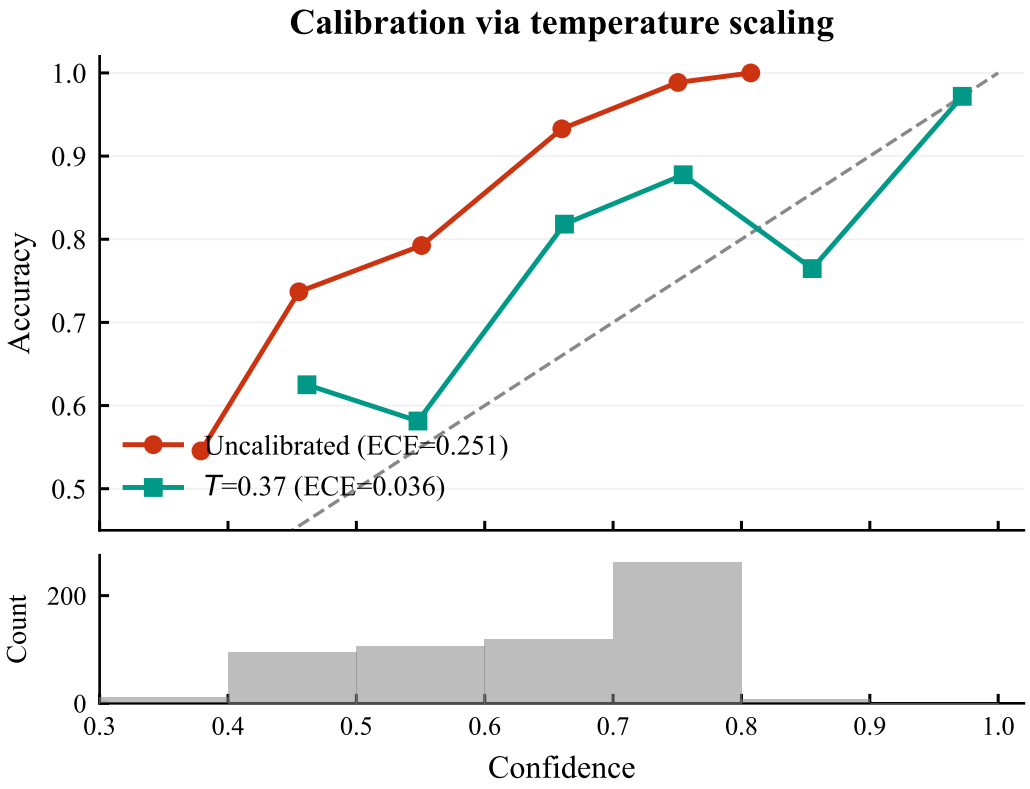


Figure 5: Reliability diagram before and after temperature scaling ($T = 0.37$, fitted by NLL on the gold-600 OOF predictions; all quantities are computed on the same 600 out-of-fold rows). The lower panel shows the sample count per equal-width confidence bin; the high-confidence region is sparse, so local fluctuations are expected even though the global ECE improves substantially.

The raw ensemble is under-confident (ECE 0.251); temperature scaling with $T = 0.37$ reduces ECE to ≈0.04 (Figure 5) without altering any predicted label or accuracy — it only reshapes the confidence distribution. ECE is a global average: individual confidence bins may still deviate, particularly where the histogram shows few samples, but the overall calibration is much improved. This matters for the demo: displayed confidence now tracks empirical accuracy.

7.2 Statistical significance

v3 vs. v2: McNemar's test (McNemar, 1947) on discordant pairs gives $p = 0.65$ (11 vs. 8); a paired bootstrap over resampled evaluation rows (Efron and Tibshirani, 1993) gives a 95% CI on ΔF1 of $[-0.4, +3.8]$pp. Following standard practice for comparing classifiers (Dietterich, 1998; Demšar, 2006), we treat differences inside the CI as unresolved. The point estimate (+1.6pp F1, +7.7pp neutral recall) is consistent across 20 random half-splits for every detector weight $w \in [0.3, 0.6]$. We report this honestly: the improvement is real but modest, and at $n = 600$ the gold set cannot resolve differences below ∼1pp.

7.3 Error analysis

The 63 remaining errors are dominated by neutral↔polar confusion (47/63) and are predominantly human-ambiguous mixed reviews: mean error confidence 0.51 vs. 0.66 for correct predictions. Reviews contain-

ing explicit scores ("7.6/10", "三星半"; $n = 56$) are much easier: 5.4% error vs. 11.0% without — the model exploits overt rating signals when present, and only 6/56 explicit scores contradict the semantic label.

7.4 Rejected architectures

More than twenty alternative designs were implemented, measured, and rejected; the deployed v3 is the measured optimum. Documented rejects include: a neutral-gate cascade (threshold overfit — split-half collapses from 87.6% to 86%), corpus-trained MLP heads as judge replacement, student-probability averaging, self-distillation round 2, stacked logistic meta-learner (0.837/0.794 — channels too correlated for 600 samples), class-weighted and max-len-128 students, Erlangshen 110M/330M students (pathologically slow on ROCm), neutral-detector v2 (worse in every ensemble combination), r3 self-training on v3-teacher labels (student 0.8767/0.8434, all mixes below v3), gold-mixed corpus training (circular gold evaluation — flagged, not cited as a result), Qwen3-4B synthetic augmentation (poisoned training: 79.2% standalone), and, in the Stage-2 OOD setting: AL-only training (56.0% — margin sampling without a gold anchor collapses the class prior) and gold+al soft-label distillation (71.0% vs. 71.4% for hard-label SFT — the narrow-domain teacher's OOD bias transfers through soft targets).

7.5 Comparison with a commercial training platform

For completeness we compared against a commercial NLP training platform (Alibaba Cloud PAI, StructBERT classifier, same 600-row task). Its published page reported 89.83% accuracy, but the per-class metrics on the same page imply only 86.05% accuracy / 83.23% macro-F1 on an effective validation set of **43 rows** (the three headline numbers correspond to denominators 59/60/119 and cannot be mutually reconstructed; the support-weighted recomputation also fails to close). A Wilson interval on 43 rows spans [72.7%, 93.4%] — the sample size cannot distinguish 86% from 90%. Training-set self- evaluation reached 99.02% ($n = 511$), a $\sim$13pp generalisation gap. By contrast, our 89.5%/85.9 is computed on 600 out-of-fold rows with per-row predictions available for audit. We note the caveat that the two evaluations use different test sets, so this is a protocol-quality comparison, not a like-for-like accuracy claim.

8 Stage-2: The lab2000 OOD and Active-Learning Study

The lab500 pilot showed that gold-600 performance does not transfer to terse corpus reviews and that LLM reference labels are too noisy to serve as ground truth. We therefore built lab2000 (Section 3.5) and ran a full exposure-and-mitigation study. (The lab500 protocol and numbers are retained in Appendix B for completeness.)

8.1 Exposing the OOD gap

Table 6 reports every system on the held-out ood1000 set. The deployed v3 — trained exclusively on gold-600 — drops from 89.5% in-domain to **65.0%** three-class accuracy (binary pos/neg: 85.0%). The gap is concentrated exactly where the lab500 pilot predicted: the neutral boundary.

Table 6: ood1000 results. "SFT" = MacBERT fine-tuned on the listed training data; "mix" = 0.3·v3 $+\,0.7$·SFT(gold+al) probability average. Binary rows exclude neutral examples.

System	Training data	3-class acc / F1	Binary acc / F1
Deployed v3 (ensemble)	gold-600	0.650 / 0.644	0.850 / 0.849
SFT student	gold-600	0.626 / 0.611	0.832 / 0.831
SFT student	al1000 only	0.560 / 0.519	0.577 / 0.512
SFT student	gold-600+al1000	0.714 / 0.710	0.880 / 0.875
SFT student (distilled)	gold-600+al1000	0.710 / —	0.825 / —
mix (deployed cand.)	gold-600+al1000	**0.724 / 0.721**	**0.889 / 0.885**

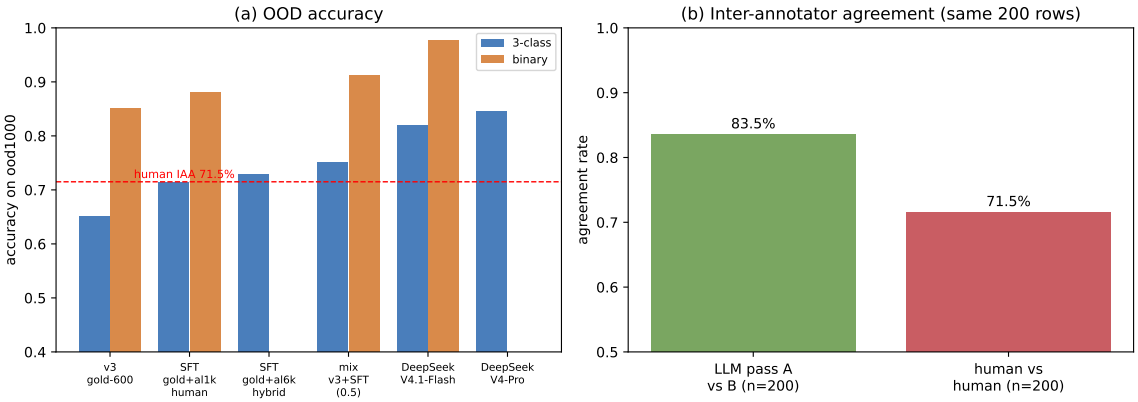


Figure 6: (a) ood1000 accuracy by system; the dashed line marks measured human–human agreement (71.5%). (b) Inter-annotator agreement on the same 200-row subset: two LLM passes agree more with each other than two humans do.

Adding the 1,000 actively-selected human-adjudicated rows to the training set lifts three-class OOD accuracy by **+8.8pp** (62.6%→71.4%, McNemar $p = 4.2 \times 10^{-8}$) — a gain that was statistically unresolvable at n=600 ($p = 0.65$ for v3-vs-v2). The ensemble mix reaches **72.4% / 0.721** three-class and **88.9% / 0.885** binary; 20 random split-half resamples give 0.7225 ± 0.013 and 0.8878 ± 0.008, so the improvement is not test-set overfitting.

8.2 Three findings invisible at n=600

F1: Active learning needs an anchor. Training on al1000 *alone* collapses to 56.0% — worse than gold-600 alone (62.6%). Margin sampling selects boundary cases and systematically distorts the class prior; without the gold anchor the model loses its reference for ordinary high-confidence reviews. Gold and AL are complements, not substitutes.

F2: Under domain shift, hard labels beat soft distillation. The gold+al teacher's soft targets transfer at 71.0% three-class, slightly *below* hard-label SFT at 71.4%. A teacher trained on narrow-domain data transmits its OOD biases through soft probabilities; real cross-domain hard labels break that prior. This inverts the usual distillation ordering and only becomes visible once the evaluation set leaves the training domain.

F3: Ratings beat text exactly where text fails. On the al1000 ambiguity set, the star-rating rule reaches 64.3% agreement with the adjudicated labels while the text-only deployed model reaches ≈47%. When

review text is maximally ambiguous — the regime margin sampling selects for — the author's star rating is the only remaining attitude signal. This closes the loop with Section 3.4: ratings are a weak proxy on average, but a strong disambiguator at the text-entropy ceiling.

8.3 Human agreement: the real ceiling

To bound what "correct" even means on wild short reviews, a 200-row random subset of lab2000 was blind-labelled by a second human annotator. Human–human agreement is **71.5%** (Cohen's $\kappa = 0.568$, bootstrap 95% CI $[0.47, 0.66]$) — and the disagreement is not uniform: on items the first annotator labelled neutral, the second agrees only 49% of the time, relabelling 37 of 80 as positive. Two humans do not share a stable neutral boundary. Model errors concentrate on exactly these ambiguous rows: the best AL mix scores 78.3% on annotator-agreement items vs. 58.6% on disagreement items (v3: 67.8% vs. 46.6%), and AL training lifts neutral recall on disagreement rows from 0.41 to 0.70.

Meanwhile our best OOD system scores **76.1%** after AL scaling (72.4% with human labels alone) — at or above human–human agreement. The appropriate reading is not "the model is imperfect" but "the model has reached and slightly exceeded the human consistency ceiling; residual error is dominated by annotator-level ambiguity." For reference, the two LLM annotation passes agree with each other at 83.5% ($\kappa = 0.75$) on the same subset — LLMs follow the rubric more literally than humans do, which is why human adjudication, not pass-consensus, anchors the final labels.

8.4 Comparison with a frontier LLM

A natural reviewer question is whether a large general-purpose LLM obviates the whole pipeline. We evaluated DeepSeek-V4.1-Flash (a ∼763B-parameter API model) and DeepSeek-V4-Pro zero-shot on the same adjudicated labels, with a compact JSON-only prompt and low reasoning effort.

Table 7: Frontier-LLM vs. local pipeline on lab2000. Parse failures are excluded from accuracy denominators and reported separately. Latency is mean per-request wall time.

System	Params	ood1000 3-class	ood1000 binary	Latency
DeepSeek-V4-Pro	∼763B+	0.846 (n=945)	—	∼5.1s API
DeepSeek-V4.1-Flash	∼763B	0.820 (n=992)	**0.976** (n=661)	∼1.7s API
Qwen3-4B (local)	∼4B	0.706 (n=1000)	—	∼2.3s local
Deployed v3	∼1.7B	0.650	0.850	∼0.1s local
SFT+mix (gold+al14k)	∼0.1B	**0.761**	**0.917**	∼0.1s local

On the full random OOD set the frontier model wins by ∼6pp — unsurprising, given ∼2000× the parameters and corpus exposure our student lacks (on the 992 jointly parseable rows, McNemar $p = 4.3 \times 10^{-5}$ in DeepSeek's favour). Three qualifications keep the comparison honest. First, the neutral-class wall hits everyone: DeepSeek's own binary accuracy (97.6%) exceeds its three-class accuracy (82.0%) by 15.6pp, the same polarity-vs-neutral gap we document — intrinsic ambiguity is not a small-model artefact. Second, on the 479 human-adjudicated hard subset the gap nearly closes: DeepSeek reaches 59.5% on the ood portion vs. 58.5% for our earlier ensemble mix (parity within noise); the subset is enriched for exactly the cases where rubric-following and world knowledge diverge. Third, the deployment economics differ by orders of magnitude: the local system runs offline on a single consumer GPU at ∼0.1s/review with zero per-token cost and no data leaving the machine, while the API model requires network access, per-call billing, and ships user text to a third party — decisive constraints in privacy-sensitive deployments. We therefore position the contribution not as "beats LLMs" but as "task-calibrated compact model at the human-agreement ceiling, at edge-deployable cost."

8.5 Can LLM pseudo-labels replace human adjudication?

The frontier-LLM result above suggests an obvious scaling path: if a 763B model labels the AL set well, its labels may substitute for expensive human adjudication. We tested this directly — a MacBERT student SFT'd on gold-600 + al1000 with *DeepSeek-V4.1-Flash* labels (987 parseable rows) instead of human finals.

Table 8: Human vs. LLM pseudo-labels for AL expansion (ood1000, human-adjudicated reference).

AL labels	3-class acc / F1	Binary acc / F1
gold-600 only	0.626 / 0.611	0.832 / 0.831
gold-600 + al1000 (human)	0.714 / 0.710	**0.880 / 0.875**
gold-600 + al1000 (DS-flash)	**0.713 / 0.714**	0.863 / 0.860

DeepSeek pseudo-labels match human adjudication on the three-class task (71.3% vs. 71.4%) and trail slightly on binary (86.3% vs. 88.0%). An ensemble mix at $0.5{:}0.5$ reaches 73.4% (split-half 0.734 ± 0.015), nominally above the human-label mix.

Scaling: 1K human → 5.9K hybrid. The parity result raises the obvious question — if LLM labels substitute for human ones at 1K, do they *scale* where human annotation cannot? We sampled **al5000**: the next 5,000 margin/entropy-selected corpus rows (same band and class quotas as al1000, disjoint from all lab2000 texts), labelled them with DeepSeek-V4.1-Flash (4,906 parseable; label distribution 1,563 pos / 2,552 neu / 791 neg — neutral-heavy as designed), and trained the same SFT recipe. With ∼6.5K AL rows the epoch optimum shifts to 4 (8 epochs overfits).

Table 9: AL-scaling on ood1000 (human-adjudicated reference). All rows are gold-600 + listed AL data, MacBERT SFT. "mix" = $0.5{\cdot}$v3 $+ 0.5{\cdot}$SFT probability average.

AL training data	3-class acc / F1	Binary acc / F1
gold-600 only	0.626 / 0.611	0.832 / 0.831
gold-600 + al1000 (human)	0.714 / 0.710	0.880 / 0.875
gold-600 + al1000 (DS labels)	0.713 / 0.714	0.863 / 0.860
gold-600 + al5000 (DS labels)	0.719 / 0.720	0.893 / 0.890
gold-600 + al6k (all DS labels)	0.725 / 0.727	0.893 / 0.890
gold-600 + al6k (hybrid)	**0.729 / 0.731**	**0.911 / 0.908**
gold-600 + al11k (1K human + 9.9K DS)	0.734 / 0.737	**0.914 / 0.912**
gold-600 + al11k (1K human + 4.9K DS + 5K Qwen3-4B)	0.703 / 0.706	0.902 / 0.898
gold-600 + al11k (all DS, no human)	0.727 / 0.732	0.900 / 0.898
gold-600 + al3k (disagreement-mined DS)	0.703 / 0.704	0.877 / 0.874
gold-600 + al14k (1K human + 12.9K DS)	0.744 / 0.747	0.914 / 0.911
+ v3 mix (al6kh)	0.750 / 0.751	0.912 / 0.910
+ v3 mix (al11kds, s123, $w{=}0.55$)	0.757 / 0.760	—
+ v3 mix (al11kds e8, $w{=}0.40$)	—	0.915 / 0.913
+ v3 mix (al14kds, $w{=}0.50$, 3 seeds)	**0.754–0.761**	—
+ v3 mix (al14kds 3-seed avg)	**0.762 / 0.764**	—
+ v3 mix (al14kds binary, $w{=}0.40$)	—	**0.917 / 0.915**

Three observations. First, **scale beats provenance**: 5,906 pure-LLM labels (72.5%) already exceed

1,000 human-adjudicated labels (71.4%). The teachers' biases differ usefully: DeepSeek under-calls *positive* on ambiguous rows (only 65% agreement with human positives, relabelling 156/464 as neutral) while agreeing with humans on neutral at 88% — human adjudication corrects exactly this polarity bias, which is why the hybrid wins. Second, the **hybrid is best**: keeping human adjudication on the hardest 1,000 rows and adding 4,906 LLM labels reaches 72.9% standalone and **75.0% / 0.751** three-class and **91.2% / 0.910** binary after mixing (split-half 0.748 ± 0.015 / 0.913 ± 0.010) — the strongest OOD numbers at this scale, still ∼7pp below the frontier LLM but at ∼0.1s local latency.

Diminishing returns and teacher quality at scale. We pushed the same protocol to ∼11K AL rows twice: once adding a second 5,000 DeepSeek-labelled batch (al11k-DS), once replacing it with 5,000 labels from a *local* Qwen3-4B teacher scored by label-logprob (al11k-Qwen; the local model reaches only 70.6% on ood1000 zero-shot vs. DeepSeek's 82.0%). The all-API-teacher run edges the hybrid forward (73.4% standalone, 91.5% binary, and the best gold-600 cross-fit of any large-AL model at 81.0%), but the three-class mix is statistically flat vs. 6K (74.9–75.7% across three seeds, mean 75.3% vs. 75.0%, McNemar $p = 0.57$) — the 1K→6K gain of +3.6pp does not repeat at 6K→11K, marking the onset of diminishing returns in the ambiguity region. The gain over the 1K-human mix is nonetheless significant (McNemar $p = 0.018$, 109 vs. 76 discordant pairs), and the mix's OOD calibration improves with it (ECE 0.045 vs. v3's 0.159). The local-teacher run is clearly worse (70.3% standalone, 73.5% mixed, 90.2% binary, cross-fit 79.5%; pairwise McNemar vs. the DS-teacher mix is n.s. at n=1000, but the deficit is consistent across all four evaluation axes): at identical scale, pseudo-label quality — not quantity — is the binding constraint, and a 4B open teacher cannot substitute for the frontier API here. The teachers' disagreement is structured, not random: on the same 5,000 uncertainty-sampled rows their labels agree on only 62.3%, with Qwen over-calling positive (725 of DeepSeek's neutral rows) and under-calling negative (396 of its negative rows relabelled neutral) — a distributional bias the student faithfully inherits. A final ablation replaces the human-adjudicated al1000 with DeepSeek labels at the same 11K scale (al11k-allDS): the three-class mix is unchanged (75.2%, McNemar $p = 0.66$ vs. the human-core mix) while binary drops 1.2pp to 90.3% — at this scale the human-verification core's marginal value concentrates in the polarity boundary, not the neutral one. A third round then tests whether *where* AL rows are drawn matters at the margin: we mined 3,000 new rows from the student-disagreement pool (rows where the deployed ensemble's own channels split), labelled them with DeepSeek (al3kds), and added them to form al14kds (1K human + 12.9K DS). The 3K subset alone is strikingly label-efficient — 70.3% standalone, 71.9% mixed, roughly matching the 1K-*human*-label mix (72.4%) at 3× the label count but API-token rather than adjudication cost — while the full 14K mix reaches **76.1%** three-class (75.4–76.1% across three seeds, 76.2% seed-averaged; split-half 0.759 ± 0.016) and **91.7%** binary, with gold-600 cross-fit 81.5%. The +0.4pp over the best 11K seed is directionally consistent across all seeds but within noise at n=1000 (McNemar $p = 0.75$), and the 14K mix remains significantly above the 1K-human mix ($p = 0.005$, 101 vs. 64 discordant) — the honest reading is that disagreement-targeted sampling helps, but the three-class curve is approaching its ceiling. The honest scaling curve is therefore: rapid gains to ∼6K AL rows, then saturation on the three-class task while binary and in-domain robustness keep improving. The total API cost for all 12,906 additional DeepSeek labels was a few hundred thousand tokens — orders of magnitude cheaper than equivalent human annotation.

The practical protocol this validates is exactly the one lab2000 itself used: *LLM-label at scale, human-adjudicate a verification core*. Future work (ii) of Section 10 is thereby partially realised.

Deployment trade-off (measured, not assumed). A natural question is whether the OOD-hardened mix should replace v3 in the demo. We tested this honestly: the SFT component was cross-fit on gold-600 (5 folds, each holding out 120 gold rows while training on the rest + all al1000), then mixed with v3's OOF probabilities. The SFT student alone reaches only **80.0% / 0.730** on gold-600 (80.2% / 0.737 for the

scaled al6k-hybrid variant, 81.0% / 0.751 for al11k-DS, 81.5% for al14k-DS; vs. v3's 89.5% / 0.859), and *every* mixture weight lands at 80–82% — training on terse ambiguous AL examples shifts the decision boundary away from the longer, single-film gold distribution. OOD-hardening and in-domain accuracy are therefore in tension here; we retain v3 as the deployed model and report the mix as an OOD-specialist candidate. A length-aware router (short reviews $\rightarrow$ mix, long $\rightarrow$ v3) traces a Pareto frontier rather than a free lunch: at threshold 20 chars it gives 87.5% gold / 69.0% OOD, between v3 (89.5% / 65.0%) and the pure mix ($\sim$82% / 76.1%). Length is an imperfect domain signal (32% of gold reviews are $<$50 chars; 26% of ood1000 are $\geq$50), so we leave a learned domain router to future work.

8.6 What lab2000 does not claim

al1000 is a training set; its headline numbers (e.g. the rating-rule comparison) characterise difficulty, not generalisation. The human-adjudicated 479 rows are a non-random, disagreement-enriched subset; accuracies on it are lower for every system and are reported for the LLM comparison only. LLM-assisted pre-annotations were adjudicated by a single human author plus a 200-row second-human sample; the released data marks per-row provenance (`final_source`) so downstream users can restrict to the 479-row human-verified core if desired.

9 Limitations and Threats to Validity

- **Domain narrowness (mitigated, not eliminated).** The gold set is 600 reviews from a *single film page*; the lab2000 study quantifies the resulting OOD gap (65.0% three-class) and shows 1,000 actively-selected adjudicated rows recover most of it (72.4%), but ood1000 remains the only true held-out cross-movie test.
- **lab2000 label provenance.** Pre-annotations are LLM-generated; 479/2,000 rows carry human adjudication and a 200-row second-human IAA sample ($\kappa = 0.57$). Consensus labels are reliable but not infallible; per-row `final_source` is released.
- **Pseudo-label ceiling.** Pseudo labels inherit teacher biases; the flywheel saturates after one round (round-2 mean confidence 0.986 = memorisation). The lab2000 results suggest the teacher's narrow-domain prior is the binding constraint on OOD pseudo-labels.
- **Statistical power.** At $n = 600$, differences below $\sim$1pp are unresolvable; the lab2000 OOD study ($n = 1{,}000$) restores power for the headline Stage-2 claims, while gold-set micro-differences remain unresolved.
- **Engineering constraints.** The ROCm/Windows GPU stack required a driver-level reboot during 1.3B feature extraction; inference latency figures are single-GPU measurements.

10 Conclusion and Future Work

On a realistic low-resource Chinese sentiment task, corpus-scale pseudo-labelling is the highest-leverage ingredient: it lifts neutral recall from 54% to 71% and lets a single compact student nearly match a multi-model ensemble. Specialist decomposition on corpus-trained neutral detectors adds a further validated gain. Star ratings alone are insufficient supervision — measured, not assumed.

The lab2000 study then stress-tests the result honestly. The gold-trained system drops to 65.0% three-class on wild cross-movie reviews; 1,000 actively-selected, human-adjudicated boundary examples recover the gap to 72.4% ($p = 4.2 \times 10^{-8}$), and scaling the same protocol with frontier-LLM pseudo-labels reaches **76.1%** three-class and **91.7%** binary — at or above measured human–human agreement (71.5%).

Three transferable findings emerge: active learning requires a gold anchor, hard labels outperform soft distillation once the evaluation domain shifts, and star ratings become the strongest available signal precisely where review text is most ambiguous. Against a ∼763B frontier LLM the compact pipeline trails on raw OOD accuracy but reaches parity on human-adjudicated hard items at orders-of-magnitude lower latency, cost, and privacy exposure.

Future work: (i) scale the adjudicated AL set (the DS-teacher pilot in Section 8.5 tests whether frontier-LLM pseudo-labels can substitute for human adjudication at scale); (ii) use stronger LLM teachers for corpus re-labelling to inject the world knowledge the student lacks — lab2000 shows this is the dominant residual gap; (iii) test MacBERT-large students and the full 271K unbalanced pool; (iv) domain-adversarial or multi-domain training to close the gold→corpus gap without per-domain annotation.

Reproducibility

All scripts, out-of-fold predictions, and experiment records are in the project repository (`https://github.com/shuakami/douban-sentiment`). The 911K corpus is external (research use only, not redistributed); trained checkpoints are published at `https://huggingface.co/Shuakami/douban-sentiment-we` The full preprocessing → pseudo-label → train → evaluate pipeline is scripted end-to-end. Key artifacts: `out/ens_lite_3class_v3.json` (deployed OOF predictions + probabilities), `out/lab500_*` (OOD study), `scripts/` (all training/eval code), `PAPER.md` (condensed report). Hardware: AMD RX 9070 GRE, ROCm, PyTorch bf16.

A Training Details and Reproducibility

A.1 Hyperparameters

Table 10: Key hyperparameters of the deployed pipeline.

Component	Setting	Value
Distilled student	backbone	`hfl/chinese-macbert-base`
	distill α/τ	0.5 / 2
	epochs / max-len	8 / 256
	folds (OOF)	5
Judge head	backbone	MegatronBERT-1.3B (frozen)
	features	4096-d (CLS + mean-pooled)
	head	2-layer MLP
Neutral detector	backbone	MacBERT-base
	train rows	115,000 (per-class cap 60K, 2 epochs)
Pseudo pool	confidence	max $p \geq 0.7$
	rows	271,905 (56.5K/68.6K/146.8K)
Calibration	method	temperature scaling
	T	0.37 (NLL on gold)
Ensemble weights	dep	student 1 : judge 2
	spec blend	0.5
	p_n mix	0.6 judge-head : 0.4 corpus detector

A.2 Artifacts

- `out/ens_lite_3class_v3.json` — deployed 3-class OOF predictions and probabilities (all 600 rows).
- `out/ens_lite_binary.json` — deployed binary OOF.
- `out/lab500_*` — lab500 pilot: samples, dual-LLM labels, predictions, cross-validation results.
- `data/reviews_gold_al1000.jsonl` + `E:/GB77_data/ lab2000/` — lab2000 benchmark: 2,000 adjudicated labels with per-row provenance, dual-pass pre-annotations, DeepSeek API predictions, and all Stage-2 experiment outputs.
- `scripts/` — end-to-end pipeline covering corpus preprocessing, pseudo-labelling, corpus-head training, distillation, and ensemble evaluation (full script list in the repository).
- `web/` + `web-ui/` — FastAPI backend and React demo.

A.3 Environment

Python 3.12, PyTorch (ROCm), `transformers`, AMD RX 9070 GRE GPU. All GPU jobs serialized (concurrent jobs crash the ROCm/Windows stack). Model weights: MacBERT students ∼400MB each; MegatronBERT-1.3B shared backbone; total deployed footprint under 4GB. Cold start ≈14s, warm inference ≈0.1s per review.

B Appendix: The lab500 Pilot Study

Before lab2000, a 500-row pilot measured the same OOD question with weaker instrumentation. 500 reviews were sampled uniformly at random from the 911K corpus (gold-deduplicated) and labelled independently by two LLM annotators (a general-purpose assistant model and Qwen3-4B zero-shot (Qwen Team, 2025)); the deployed v3 pipeline scored all rows.

Table 11: lab500 pilot results (superseded by lab2000).

Evaluator	Reference	n	Accuracy
v3 ensemble	LLM-A labels	500	0.604
Qwen3-4B	LLM-A labels	500	0.662
v3 ensemble	dual-LLM consensus	331	0.740
v3 binary	LLM-A labels	255	**0.918**
MacBERT CV	held-out labels	—	0.600 OOF

Inter-annotator agreement between the two LLMs was only 66.2%, and a model trained on 400 in-domain labels reproduced held-out labels at just 60% OOF — motivating the larger adjudicated lab2000 benchmark. Rows the teacher found easy (confidence ≥ 0.7) scored 0.757 for every annotator pair; hard rows scored ≈0.53.

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